

CICA: Coupling Confidence-Aware Pretraining with Confidence-Informed Attention for Robust Multimodal Sentiment Analysis

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Abstract

Multimodal sentiment analysis requires integrating language, visual, and acoustic cues, yet these modalities are often noisy, incomplete, or contradictory, making fusion unreliable. Most existing methods assume uniformly trustworthy modalities and thus degrade when signals conflict. To address this, we propose CICA, a framework that couples Confidence-Aware Pretraining with Confidence-Informed Attention. In pretraining, each modality encoder learns to estimate the reliability of its own representation, producing both embeddings and confidence scores. These scores then guide a confidence-informed attention mechanism, which strengthens contributions from reliable modalities while suppressing noisy or conflicting ones, enabling adaptive fusion under varying signal conditions. CICA achieves state-of-the-art performance across four major benchmarks on MOSI, MOSEI, CH-SIMS, and CH-SIMSV2. It achieves MAE 0.630 and Corr 0.855 on MOSI, and MAE 0.489 and Corr 0.856 on MOSEI, significantly surpassing prior methods. Consistent improvements are also observed across Acc-7, Acc-2, and F1 metrics. Under noisy and missing-modality conditions, CICA maintains significantly more stable performance, indicating improved robustness and interpretability.

1. Introduction

Multimodal sentiment analysis (MSA) aims to infer human affective states by jointly modeling language, visual, and acoustic signals. As these modalities are inherently interdependent in human communication, their integration plays a crucial role in building reliable perception systems for emotion understanding [2, 21]. However, real-world mul-

MOSEI Dataset



Utterance: I have applied it to behaviors tho, the best example I could give.....

- Text: Semantically Neutral .
- Audio: Conveys Strong Negative sentiment (a frustrated tone)
- Video: Expresses Strong Positive emotion (a broad smile)

Ground Truth : Labeled as Strongly Negative (-1.33)

Figure 1. An illustrative case from MOSEI showing modality misalignment. The visual cue (smile) conveys strong positivity but is misaligned with a negative acoustic signal (frustrated tone) and neutral text. The ground truth sentiment follows the acoustic modality (-1.33), yet traditional models are easily misled by the visual cue.

timodal data are rarely clean or temporally aligned. They often contain inconsistent cues, asynchronous signals, and, most critically, **modality conflicts**, which severely degrade model robustness.

As shown in Fig. 1, even advanced multimodal Transformers can be easily confused by cross-modal inconsistencies. While deep fusion architectures are effective at correlating multimodal features, they typically rely on uniform attention mechanisms that treat all modalities as equally reliable. These models thus lack an internal mechanism to assess which modality should dominate under conflict—leading to erratic behavior when confronted with noise, temporal drift, or missing information.

We argue that robustness in multimodal systems requires a new paradigm: fusion must be conditioned on each modality’s perceived reliability. In other words, the model must first measure how trustworthy each modality is for a given sample before deciding how to combine them. This motivates our design of a coupled “perceive-and-decide” framework.

To this end, we propose CICA—Coupling Confidence-

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Aware Pretraining with Confidence-Informed Attention. CICA establishes a direct coupling between self-assessed confidence and fusion-time decision-making, implemented in two coordinated stages: **i. Confidence-Aware Pretraining (CAP).** Each unimodal encoder ($\mathbf{E}_m, m \in \{T, V, A\}$) is trained not only to extract representations $\mathbf{H}_m$ but also to estimate a sample-specific confidence score s_m , reflecting the reliability of its own output for the target task. Rather than relying solely on feature quality, CAP employs an adaptive multi-task pretraining objective (detailed in Sec. 3.1) that explicitly calibrates the encoder’s uncertainty. **ii. Confidence-Informed Fusion (CIF).** In the fusion stage, the confidence signals estimated during pretraining are used to regulate cross-modal integration. The core of this stage is the **Confidence-Informed Attention (CIA)** mechanism, which captures intrinsic structural relations in the multimodal feature space—such as token-level saliency ρ_m and local contextual dependencies $\mathbf{N}_m$ —as elaborated in Sec. 3.2.

Subsequently, the reliability estimates (s_m, u_m) produced by CAP are applied as continuous modulation factors to the attention outputs. This design links the perception of signal reliability to the decision-level fusion weights, ensuring that the model dynamically allocates trust to modalities according to their measured confidence. To prevent “modality collapse”—where one modality dominates fusion—CICA further introduces a Mutual-information Contrastive Preservation (MCP) objective (Sec. 3.3), which regularizes the fused representation to retain unique modality contributions. We validate CICA across four major benchmarks—MOSI, MOSEI, CH-SIMS, and CH-SIMSV2—achieving new state-of-the-art results, including MAE 0.630 / Corr 0.855 on MOSI and MAE 0.489 / Corr 0.856 on MOSEI. Moreover, CICA exhibits substantially improved robustness under synthetic noise and modality-missing evaluations (Sec. 4.5), highlighting the benefit of coupling perception and decision mechanisms. Our work makes the following key contributions:

- i. We propose **CICA**, a unified “perceive-and-decide” framework that couples confidence perception with fusion-time decision-making for robust multimodal sentiment analysis.
- ii. We introduce CAP, which trains unimodal encoders to quantify their own reliability through sample-specific confidence estimation.
- iii. It uses the reliability signals (s_m, u_m) from CAP to continuously modulate the outputs of this CIA mechanism, which (as described in Sec. 3) first perceives intrinsic feature quality before the reliability signals are applied, thereby explicitly linking perception with the final decision.
- iv. We achieve SOTA performance and demonstrate superior robustness under noise and missing-modality con-

ditions (as described in Sec. 4.5), validating the efficacy of confidence-driven multimodal reasoning.

2. Related Work

2.1. Multimodal Fusion Architectures

Integrating heterogeneous data streams is a central challenge in MSA. Early tensor-based methods, such as TFN [2] and LMF [11], capture high-order cross-modal interactions. Transformer-based attention models, *e.g.* MuT [14], have since become standard.

However, uniform processing of modalities makes these methods vulnerable to conflicts (*e.g.* sarcasm). Asymmetric or guided fusion approaches [5, 17, 18] aim to mitigate modality bias. Some assume text dominance (ALMT [24]), while others use dynamic guidance (KuDA [3], CLGSI [19]), often relying on external knowledge or lacking a principled, sample-specific reliability measure.

2.2. Reliability and Uncertainty Modeling

A parallel line of work quantifies per-modality reliability to overcome static fusion limitations, forming the basis of our “perceive-and-decide” paradigm. Methods such as ULFHUC [12] use Evidential Deep Learning [1] or Bayesian approximations [4] to estimate uncertainty, typically applied post-hoc at the decision layer.

Previous studies have applied the Three-Way Decision theory [20] to design multimodal fusion strategies. Typical implementations (*e.g.* 3WD-DRT [9]) predict a single modality confidence score (s_m), which is then used to partition each modality into discrete decision regions—such as acceptance, deferment, or rejection—with separate scaling functions applied to each region.

CICA introduces two fundamental advances. First, in the perception phase (CAP), instead of relying on a single confidence signal, CICA jointly models both adaptive confidence and task-specific uncertainty, allowing it to disentangle epistemic and aleatoric reliability (see Sec. 3.1). Second, in the decision phase (CIA), these dual reliability cues are coupled through a selective gating mechanism that amplifies a modality only when it is both confident and consistent. Unlike 3WD-DRT and similar works that rely solely on confidence-based scaling, this stricter reliability gating leads to more stable and interpretable fusion.

2.3. Representation and Information Preservation

Other works enhance representations (*e.g.* MISA [7], SelfMM [22]) or explore efficient architectures (*e.g.* CubeMLP [13] and TMBL [8]).

Most relevant to our auxiliary objective is Mutual Information Maximization [6]. We use this concept to prevent “modality collapse.” We introduce a MCP loss (Sec. 3.3), inspired by Contrastive Predictive Coding [15]. This en-

sure our CIA fuser retains sufficient information from all sources, even when suppressing a deceptive modality.

3. Methodology

Our framework, **CICA**, addresses modality conflicts and noise by building a “self-aware” system. The central idea is to inform the fusion (decision) module with the encoder’s self-assessed signal quality (perception). As shown in Fig. 2, the pipeline has two phases. **Phase 1, CAP** (Stages 1–2), trains each unimodal encoder $\mathbf{E}_m$ to generate representations and sample-specific reliability metrics (confidence s_m , uncertainty u_m). These metrics provide self-awareness of signal quality. **Phase 2, CIF** (Stages 3–4), freezes the pretrained encoders. Then a fusion module $\mathbf{F}$ is trained, taking advantage of the reliability signals (s_m, u_m) for adaptive multimodal integration.

3.1. Phase 1: CAP

The CAP phase trains each unimodal encoder $\mathbf{E}_m$ ($m \in \{T, V, A\}$) to output representations and reliability indicators via $(\mathbf{H}_m, s_m, u_m) = \mathbf{E}_m(\mathbf{X}_m)$.

As illustrated in Fig. 2 (Stage 2), each encoder $\mathbf{E}_m$ uses a Transformer backbone [16]. A MixDomainAdapter disentangles task-specific information ($\mathbf{h}_{\text{spci}}$) from the final layer and domain-related features ($\mathbf{h}_{\text{domain}}$) from intermediate layers. These are concatenated:

$$\mathbf{H}_m = \text{Concat}(\mathbf{h}_{\text{spci}}, \mathbf{h}_{\text{domain}}). \quad (1)$$

Moreover, each encoder is pretrained with three auxiliary heads under a joint objective:

$$\mathcal{L}_{\text{CAP}} = \mathcal{L}_{\text{task_pre}} + \lambda_{\text{CA}} \mathcal{L}_{\text{CA}} + \lambda_{\text{uncert}} \mathcal{L}_{\text{uncert}}. \quad (2)$$

The task head $\mathbf{C}_{\text{pred}}$ predicts unimodal sentiment $\hat{y}_m$ using an L2 loss (MSE):

$$\mathcal{L}_{\text{task_pre}} = \frac{1}{B} \sum_{i=1}^B (y_{m,i} - \hat{y}_{m,i})^2. \quad (3)$$

The confidence head $\mathbf{C}_{\text{CA}}$ predicts a scalar confidence score $s_m \in [0, 1]$. Standard losses are poorly suited for this task: classification losses (e.g. BCE) incorrectly frame it as a binary problem, while static regression losses (e.g. $\mathcal{L} = (s_m - 1)^2$) are too rigid. We therefore propose an Adaptive CA loss, $\mathcal{L}_{\text{CA}}$, which serves as a *learnable calibration function*. It uses learnable parameters θ_{CA} to adaptively define reliability boundaries, enabling the encoder to function as an adaptively calibrated perceiver. The full formulation, parameterization, and justification are detailed in **Appendix H.1**. These parameters $\theta_{\text{CA}} = \{\hat{\alpha}, \hat{\beta}, \hat{w}_{\text{high}}, \hat{w}_{\text{mid}}, \hat{w}_{\text{low}}\}$ define two boundary points ($0 < \beta < \alpha < 1$) and three corresponding penalty weights

($w_{\text{high}}, w_{\text{mid}}, w_{\text{low}}$). The boundaries are constrained via sigmoid functions:

$$\beta = \sigma(\hat{\beta}), \quad \alpha = \beta + (1 - \beta) \cdot \sigma(\hat{\alpha}), \quad (4)$$

while all weights are ensured to remain positive via the Soft-plus operation. The parameters θ_{CA} are trained jointly with the rest of the encoder with the main objective $\mathcal{L}_{\text{CAP}}$. The weighting function $W(s_m)$ applies a piecewise penalty:

$$W(s_m, \theta_{\text{CA}}) = \begin{cases} w_{\text{low}} & \text{if } s_m \geq \alpha \\ w_{\text{mid}} & \text{if } \beta \leq s_m < \alpha \\ w_{\text{high}} & \text{if } s_m < \beta \end{cases} \quad (5)$$

Finally, the objective $\mathcal{L}_{\text{CA}}$ is formulated as a weighted loss L_2 that encourages high confidence:

$$\mathcal{L}_{\text{CA}} = \frac{1}{B} \sum_{i=1}^B W(s_{m,i}, \theta_{\text{CA}}) \cdot (s_{m,i} - 1)^2. \quad (6)$$

The uncertainty head $\mathbf{C}_{\text{uncert}}$ predicts the bounded absolute error of the task head, $u_m \in [0, 1]$, with a detached target:

$$u_{\text{target}} = \tanh(|y_m - \hat{y}_m^{\text{detach}}|), \quad (7)$$

and optimized by:

$$\mathcal{L}_{\text{uncert}} = \frac{1}{B} \sum_{i=1}^B (u_{m,i} - u_{\text{target},i})^2. \quad (8)$$

After pretraining, the encoders $\mathbf{E}_m$ are frozen, providing features and reliability signals (s_m, u_m) for fusion.

3.2. Phase 2: CIF

Our decision module, **CIF**, implements the “perceive-and-decide” paradigm through a decoupled two-stage process. It first uses our novel Confidence-Informed Attention (CIA) mechanism to perceive intrinsic feature structure, and then uses CAP signals to apply extrinsic reliability modulation. Our core premise is that standard attention ($Q \cdot K^\top$) only models relevance, but ignores the intrinsic quality of the key $\mathbf{K}_m$ itself. Our confidence-informed analogy is a deliberate mechanism to model this missing quality signal. Given a query $\mathbf{Q} = \mathbf{Z}^{(l-1)}$ and unimodal key-value pairs $(\mathbf{K}_m, \mathbf{V}_m)$, the fusion process is as follows.

To complement the standard relevance score ($Q \cdot K^\top$), we introduce S_{mod} , an auxiliary *Intrinsic Structure Modulator*. This term models the key’s $\mathbf{K}_m$ intrinsic quality by capturing local dependencies ($\mathbf{N}_m$) and token-level saliency (ρ_m). Our ablation study (Variant E, Tab. 3) confirms this provides a consistent performance boost. The detailed formulation and its implementation are unified in Appendix A, which presents the full forward pass Algorithm 2. This

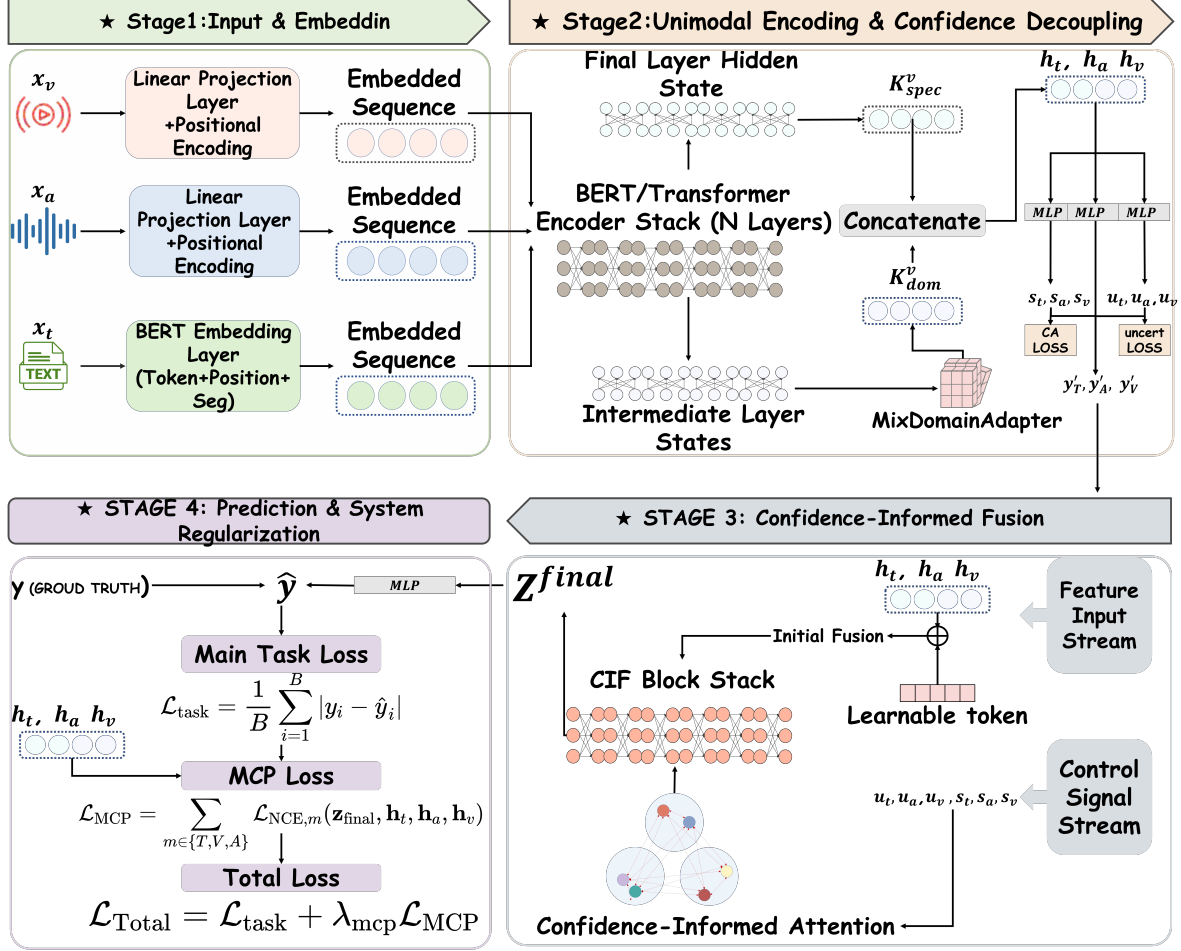


Figure 2. The architecture of our CICA framework, which explicitly couples perception and decision. **Phase 1 (CAP):** Stage 2 encoders are trained as “perceivers” in the CAP stage using $\mathcal{L}_{\text{task-pre}}$, $\mathcal{L}_{\text{CA}}$, and $\mathcal{L}_{\text{uncert}}$ to produce features ($\mathbf{H}_m$) and reliability metrics (s_m, u_m). **Phase 2 (CIF):** The perceived metrics (s_m, u_m) are provided to Stage 3 as a control signal stream that guides the CIF stage. Stage 4 applies $\mathcal{L}_{\text{MCP}}$ to regularize the final prediction ($\hat{y}$) and prevent modality collapse.

term is added inside the attention mechanism to produce a structure-aware representation $\mathbf{z}_{\text{struct},m}$:

$$\mathbf{z}_{\text{struct},m} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_m^\top}{\sqrt{d_k}} + S_{\text{mod}}\right) \mathbf{V}_m. \quad (9)$$

The final coupling between pretraining and fusion is realized within the CIF module. The reliability cues (s_m, u_m) estimated by the frozen CAP encoders are used to modulate the output of the CIA mechanism. Two complementary reliability signals are employed: (i) s_m denotes the adaptive confidence score generated by the C_{CA} head and optimized via $\mathcal{L}_{\text{CA}}$, representing the model’s self-assessed reliability for each modality. (ii) u_m represents the predicted absolute error produced by the C_{uncert} head, optimized via $\mathcal{L}_{\text{uncert}}$ to explicitly estimate the expected prediction error of the unimodal task head for the given sample.

They are combined through a projection function $g(\cdot)$ to

produce a unified reliability score:

$$r_m = g(s_m, u_m) = \text{ReLU}(1 + s_m - u_m) \quad (10)$$

This design encourages high contribution only when confidence is strong and uncertainty is low. The resulting reliability r_m then continuously modulates the feature response from the perception stage:

$$\mathbf{z}_{\text{CIF},m} = \mathbf{z}_{\text{struct},m} \times r_m \quad (11)$$

where r_m is broadcast across sequence and feature dimensions. As a result, modalities assessed as unreliable ($r_m \approx 0$) contribute less to the joint representation, while consistent signals are naturally emphasized, as observed in the qualitative analysis (Sec. 4.6).

3.3. Final Objective

After N CIF layers, the final fusion state $\mathbf{z}_{\text{final}} = \mathbf{z}_{\text{CIF}}^{(0)}$ is used by the classifier to predict the overall sentiment $\hat{y}$. The main objective is the L1 loss:

$$\mathcal{L}_{\text{task}} = \frac{1}{B} \sum_{i=1}^B |y_i - \hat{y}_i|. \quad (12)$$

To prevent modality collapse, we introduce a MCP loss. Following Noise-Contrastive Estimation (NCE) [15], $\mathcal{L}_{\text{MCP}}$ ensures the fused $\mathbf{z}_{\text{final}}$ retains correspondence with each unimodal source $\mathbf{h}_m = \text{MeanPool}(\mathbf{H}_m)$:

$$\mathcal{L}_{\text{MCP}} = \sum_{m \in \{T, V, A\}} \mathcal{L}_{\text{NCE}, m}(\mathbf{z}_{\text{final}}, \mathbf{h}_m). \quad (13)$$

The overall training objective for fine-tuning is:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{mcp}} \mathcal{L}_{\text{MCP}}. \quad (14)$$

4. Experiments

4.1. Datasets and Evaluation Metrics

We evaluate CICA on four widely-used multimodal sentiment analysis benchmarks: CH-SIMS [21], CH-SIMSV2 [10], MOSI [23], and MOSEI [2]. As detailed in Tab. 1, these datasets cover both Chinese and English. Notably, CH-SIMSV2 is a significant upgrade to CH-SIMS (4403 vs. 2281 samples), with finer-grained 5-class labels (vs. 3-class) and stricter modality alignment protocols [10]. These additions make it a more challenging benchmark for assessing robustness against modality misalignment. We adhere to the official train/validation/test splits for all datasets. For evaluation, we follow established practices [7, 22, 24]. We report regression metrics (MAE $\downarrow$, Corr $\uparrow$) and classification metrics (F1 $\uparrow$, Acc-2/3/7 $\uparrow$). The precise evaluation protocols for all metrics are detailed in Appendix B.

4.2. Baselines and Training Setup

We compare CICA against a range of state-of-the-art baselines, organized into two groups. The first group includes classic and Transformer-based fusion models, which focus on integrating multimodal features: TFN [2], LMF [11], MuLT [14], MISA [7], SelfMM [22], CubeMLP [13], and TMBL [8]. The second group consists of guided or asymmetric fusion models, designed to balance information or adaptively mitigate modality bias. This includes text-bias correction models such as MMIM [6], BBFN [5], CENet [18], TETFN [17], ALMT [24], as well as recent dynamic guidance strategies such as KuDA [3] and CLGSI [19].

All experiments are implemented in PyTorch. Training follows a two-phase procedure: CAP pretraining and CIF finetuning. To keep the main text concise, we report only the task-critical hyperparameter—the MCP loss

weight—set to $\lambda_{\text{mcp}} = 0.1$ based on validation performance. Full training configurations, including optimizer settings, GPU specifications, epochs, learning rates, detailed settings and a sensitivity analysis validating our choices are provided in Appendices C and D, respectively.

Table 1. Statistics of the four datasets used in our experiments.

Dataset	Train	Valid	Test	Total	Lang.
CH-SIMS	1368	456	457	2281	CH.
CH-SIMSV2	2722	647	1034	4403	CH.
MOSI	1284	229	686	2199	EN.
MOSEI	16326	1871	4659	22856	EN.

4.3. Main Results

We present our main experimental results in Tab. 2, comparing CICA with a comprehensive suite of SOTA baselines. The results clearly and consistently demonstrate that our framework establishes new SOTA performance across all four benchmarks. On the highly-competitive MOSI benchmark, CICA achieves a new SOTA MAE of 0.630 and a Correlation of 0.855. This represents a massive improvement over the best prior baseline, with a $\downarrow 0.071$ drop in MAE and a $\uparrow 0.060$ gain in Correlation. This dominance is mirrored on the large-scale MOSEI dataset (MAE 0.489, Corr 0.856). The performance gains are even more pronounced on the CH-SIMS and CH-SIMSV2 datasets, which are known for their complex interactions. Notably, CICA achieves a remarkable +0.127 Correlation gain on CH-SIMS and a +4.35 Acc-3 gain on CH-SIMSV2 compared to the strongest baseline. This consistent sweep across all benchmarks validates the effectiveness of our CICA framework. The subsequent sections will demonstrate, through ablation and robustness analysis, that this significant performance gain stems directly from our proposed “perceive-and-decide” mechanism.

4.4. Ablation Study

Tab. 3 reports the ablation results of CICA on the primary MOSI and MOSEI benchmarks. The extensive ablation results on the CH-SIMS and CH-SIMSV2 datasets are provided in the supplementary material. As shown in Tab. 3, the full model consistently outperforms all variants. Removing CAP (A) or CIF (B) causes the largest drops (e.g., MOSI Corr $\downarrow 0.064/\downarrow 0.043$, MOSEI F1 (non-0) $\downarrow 6.97/\downarrow 6.26$), confirming the importance of confidence-aware pretraining and the proposed attention mechanism. Disabling explicit CAP-CIF coupling (D) still degrades performance (e.g., MOSI Corr $\downarrow 0.024$), indicating that independent modules alone are insufficient. Removing MCP (C) or the intrinsic term S_{mod} (E) results in minor yet consistent declines, suggesting they support cross-modal balance

Table 2. Main results on four datasets. We compare our CICA framework against best-performing baseline. The Δ row shows the improvement of our CICA over the best-performing baseline. **Bold** indicates the best performance and underline indicates the second-best model. For MOSI/MOSEI, Acc-2 and F1 metrics are reported as *Negative vs. Non-negative (has-0) / Negative vs. Positive (non-0)*.

Method	MOSI					MOSEI				
	MAE ↓	Corr ↑	Acc-7 ↑	Acc-2 ↑	F1 ↑	MAE ↓	Corr ↑	Acc-7 ↑	Acc-2 ↑	F1 ↑
TFN [2]	0.947	0.673	34.46	77.99/79.08	77.95/79.11	0.572	0.714	51.60	78.50/81.89	78.96/81.74
LMF [11]	0.950	0.651	33.82	77.90/79.18	77.80/79.15	0.576	0.717	51.59	80.54/83.48	80.94/83.36
MuT [14]	0.879	0.702	36.91	79.71/80.98	79.63/80.95	0.559	0.733	52.84	81.15/84.63	81.56/84.52
MISA [7]	0.776	0.778	41.37	81.84/83.54	81.82/83.58	0.557	0.751	52.05	80.67/84.67	81.12/84.66
BBFN [5]	0.796	0.744	43.88	80.32/82.47	80.21/82.44	0.545	0.760	52.88	82.87/85.73	83.13/85.56
MMIM [6]	0.744	0.780	44.75	82.51/84.30	82.38/84.23	0.550	0.761	51.88	83.75/85.42	83.93/85.26
Self-MM [22]	0.708	0.796	46.67	83.44/85.46	83.36/85.43	0.531	0.764	<u>53.87</u>	83.76/85.15	83.82/84.90
CubeMLP [13]	0.755	0.772	43.44	80.76/82.32	81.77/84.23	0.537	0.761	53.35	82.36/85.23	82.61/85.04
ALMT [24]	0.712	0.793	46.79	83.97/85.82	84.05/85.86	0.530	0.774	53.62	81.54/85.99	81.05/86.05
KuDA [3]	0.705	<u>0.795</u>	47.08	84.40/86.43	84.48/86.46	0.529	0.776	52.89	83.26/86.46	82.97/86.59
3WD-DRT [9]	<u>0.701</u>	0.794	<u>48.42</u>	<u>83.41/86.51</u>	<u>85.28/87.08</u>	<u>0.518</u>	<u>0.779</u>	53.83	<u>84.19/87.33</u>	<u>83.48/88.26</u>
Our CICA	0.630	0.855	49.56	88.19/90.24	88.14/90.23	0.489	0.856	55.29	84.72/90.18	85.15/90.16
Δ	$\downarrow 0.071$	$\uparrow 0.060$	$\uparrow 1.14$	$\uparrow 3.79/\uparrow 3.73$	$\uparrow 2.86/\uparrow 3.15$	$\downarrow 0.029$	$\uparrow 0.077$	$\uparrow 1.42$	$\uparrow 0.53/\uparrow 2.85$	$\uparrow 1.67/\uparrow 1.90$

Method	CH-SIMS					CH-SIMsv2				
	MAE ↓	Corr ↑	Acc-3 ↑	Acc-2 ↑	F1 ↑	MAE ↓	Corr ↑	Acc-3 ↑	Acc-2 ↑	F1 ↑
TFN [2]	0.432	0.591	65.12	78.38	78.62	0.303	0.707	72.21	80.14	80.14
LMF [11]	0.441	0.576	64.68	77.77	77.88	0.367	0.557	64.90	74.18	73.88
MuT [14]	0.453	0.564	64.77	78.56	79.66	0.291	0.738	73.19	80.68	80.73
BBFN [5]	0.430	0.564	61.05	78.12	77.88	0.300	0.708	71.47	78.53	78.41
Self-MM [22]	0.425	0.595	65.47	80.04	80.44	0.311	0.695	72.61	79.69	79.76
CubeMLP [13]	0.419	0.593	65.86	77.68	77.59	0.334	0.648	71.95	78.53	78.53
CENet [18]	0.471	0.534	62.58	77.90	77.53	0.310	0.699	73.10	79.56	79.63
TETFN [17]	0.420	0.577	63.24	81.18	80.24	0.310	0.695	73.65	79.73	79.81
ALMT [24]	0.408	0.594	65.86	78.77	78.71	0.308	0.700	71.86	79.59	79.51
TMBL [8]	0.429	0.592	65.43	79.12	78.75	0.313	0.706	73.02	80.46	80.36
KuDA [3]	0.408	0.613	66.52	80.74	80.71	<u>0.271</u>	0.759	<u>76.21</u>	82.11	82.04
3WD-DRT [9]	<u>0.408</u>	<u>0.627</u>	<u>66.64</u>	<u>81.37</u>	<u>81.91</u>	0.276	<u>0.764</u>	75.86	<u>83.05</u>	<u>84.01</u>
Our CICA	0.378	0.754	76.37	86.00	85.68	0.245	0.842	80.56	85.98	85.89
Δ	$\downarrow 0.030$	$\uparrow 0.127$	$\uparrow 9.73$	$\uparrow 4.63$	$\uparrow 3.77$	$\downarrow 0.026$	$\uparrow 0.078$	$\uparrow 4.35$	$\uparrow 2.93$	$\uparrow 1.88$

and attention stability. These trends are equally consistent across the Chinese datasets (detailed in Sec. E), demonstrating that CAP, CIF, and their coupling drive the majority of performance gains, while MCP and S_{mod} provide complementary robustness.

4.5. Robustness Analysis

We evaluate CICA’s robustness to noisy or missing signals via two tests: (i) injecting Gaussian noise into non-textual modalities and (ii) masking modalities at inference.

To emulate sensor degradation, we add Gaussian noise (variance 0.2–0.8) to Audio (A), Video (V), or both. Fig. 3 shows a smooth performance decay. On MOSEI, correlation drops only slightly under mild audio noise (0.856 $\rightarrow$ 0.843), indicating that CAP detects corrupted inputs and CIF down-weights them. At higher noise levels, CICA dis-

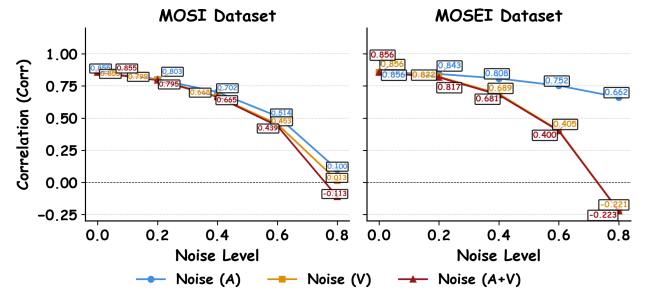


Figure 3. Robustness to modality noise on MOSI and MOSEI. Correlation (Corr) is plotted against Gaussian noise levels (0–0.8) applied to Audio (A), Video (V), or both (A+V).

plays modality-specific sensitivity: on MOSI, heavy video

Table 3. Ablation study of the CICA framework on MOSI and MOSEI datasets. Columns represent ablated variants: **(Full)** complete model; **(A)** w/o CAP; **(B)** w/o CIF; **(C)** w/o MCP; **(D)** w/o Coupling; **(E)** w/o S_{mod} . Δ denotes the absolute performance drop. Best results are in **bold**. Note: Full ablation results, including CH-SIMS and CH-SIMSv2, are provided in the supplementary material.

Metric	CICA (Full)	(A) w/o CAP	(B) w/o CIF	(C) w/o MCP	(D) w/o Coupling	(E) w/o S_{mod}
Dataset: MOSI						
MAE ↓	0.630	0.712 (↑0.089)	0.689 (↑0.066)	0.635 (↑0.012)	0.658 (↑0.035)	0.641 (↑0.018)
Corr ↑	0.855	0.791 (↓0.064)	0.812 (↓0.043)	0.847 (↓0.008)	0.831 (↓0.024)	0.840 (↓0.015)
Acc-7 ↑	49.56	46.47 (↓3.09)	47.52 (↓2.04)	49.13 (↓0.43)	48.25 (↓1.31)	48.82 (↓0.74)
Acc-2 (has-0) ↑	88.19	85.12 (↓3.07)	86.03 (↓2.16)	87.84 (↓0.35)	87.08 (↓1.11)	87.55 (↓0.64)
Acc-2 (non-0) ↑	90.24	86.88 (↓3.36)	87.90 (↓2.34)	89.51 (↓0.73)	88.73 (↓1.51)	89.30 (↓0.94)
F1 (has-0) ↑	88.14	84.03 (↓4.11)	85.21 (↓2.93)	87.05 (↓1.09)	86.53 (↓1.61)	87.48 (↓0.66)
F1 (non-0) ↑	90.23	85.05 (↓5.18)	86.13 (↓4.10)	87.66 (↓2.57)	87.02 (↓3.21)	89.15 (↓1.08)
Dataset: MOSEI						
MAE ↓	0.489	0.524 (↑0.035)	0.511 (↑0.022)	0.496 (↑0.007)	0.504 (↑0.015)	0.499 (↑0.010)
Corr ↑	0.856	0.775 (↓0.081)	0.793 (↓0.063)	0.845 (↓0.011)	0.820 (↓0.036)	0.836 (↓0.020)
Acc-7 ↑	55.29	52.81 (↓2.48)	53.54 (↓1.75)	54.82 (↓0.47)	54.17 (↓1.12)	54.50 (↓0.79)
Acc-2 (has-0) ↑	84.72	82.04 (↓2.68)	82.87 (↓1.85)	84.06 (↓0.66)	83.51 (↓1.21)	83.95 (↓0.77)
Acc-2 (non-0) ↑	90.18	87.51 (↓2.67)	88.04 (↓2.14)	89.42 (↓0.76)	88.91 (↓1.27)	89.43 (↓0.75)
F1 (has-0) ↑	85.15	82.52 (↓2.63)	83.10 (↓2.05)	84.11 (↓1.04)	83.82 (↓1.33)	84.35 (↓0.80)
F1 (non-0) ↑	90.16	83.19 (↓6.97)	83.90 (↓6.26)	84.71 (↓5.45)	84.13 (↓6.03)	88.05 (↓2.11)

noise (0.8) reduces correlation to 0.013, while audio noise at the same level still maintains 0.100, showing that the model localizes and suppresses unstable streams. Tab. 4 reveals three patterns. (i) Removing Text (T) leads to the largest degradation (MOSI: MAE 0.630 $\rightarrow$ 1.442, Corr 0.855 $\rightarrow$ 0.247), confirming its dominance. (ii) Removing non-textual signals has minimal effect. On MOSEI, T+A (no V) performs nearly the same as T+V+A (MAE 0.483 vs. 0.489), showing that CAP flags the missing modality ($s_V = 0$) and CIF adjusts its fusion. (iii) The text-only encoder is competitive and even slightly outperforms full T+V+A on MOSI (Corr 0.857 vs. 0.855), reflecting strong unimodal generalization.

This MOSI result is consistent with our “perceive-and-decide” design rather than contradicting multimodal fusion. MOSI is highly text-dominant (Appendix F), so 0.857 serves as the effective T-only upper bound. CAP assigns near-zero reliability to V/A ($s_m \approx 0$, $u_m \approx 1$), and CIF suppresses them accordingly. The 0.002 gap to T+V+A mainly reflects the mild fusion regularization from $\mathcal{L}_{\text{MCP}}$, which prevents modality collapse and preserves multimodal discriminability needed for datasets where V/A matter (e.g., MOSEI, CH-SIMS). This cost is negligible given the cross-benchmark gains.

Importantly, CICA’s advantage lies not in beating its own T-only variant but in consistently surpassing all SOTA baselines. Prior fusion models (Tab. 2) degrade sharply under conflicting or missing modalities, whereas CICA remains stable by gating unreliable signals—a trend also observed on CH-SIMS and CH-SIMSv2 (Appendix F). Thus, the internal 0.002 gap is insignificant relative to CICA’s large

improvements over existing architectures in both full- and missing-modality settings.

Table 4. Robustness on MOSI and MOSEI under different modality combinations. T+V+A denotes the full model.

Modalities	MOSI			MOSEI		
	MAE↓	Corr↑	F1↑	MAE↓	Corr↑	F1↑
T+V+A	0.630	0.855	88.14	0.489	0.856	85.15
T+V	0.663	0.839	85.53	0.540	0.838	84.61
T+A	0.695	0.840	88.06	0.483	0.850	85.71
V+A	1.442	0.247	35.99	0.811	0.250	64.51
T	0.615	0.857	86.58	0.510	0.845	85.25
V	1.435	0.094	33.18	0.811	0.232	66.00
A	1.433	0.254	40.17	0.839	0.124	57.39

4.6. Case Study: The Perceive-and-Decide Mechanism

To examine how CICA handles cross-modal inconsistency, we analyze a representative misaligned example from Fig. 1: a smiling face, neutral text, and strongly negative vocal tone ($y = -1.333$), representing typical multimodal conflict. A conventional fusion model, lacking self-assessment, over-relies on the visually dominant signal and predicts incorrectly. CICA, in contrast, resolves this via its “perceive-and-decide” process (Fig. 4). During perception, CAP encoders estimate modality confidence, identifying the visual stream as unreliable ($s_V = 0.12$, $u_V = 0.89$) and the audio stream as highly reliable ($s_A = 0.91$, $u_A = 0.15$).

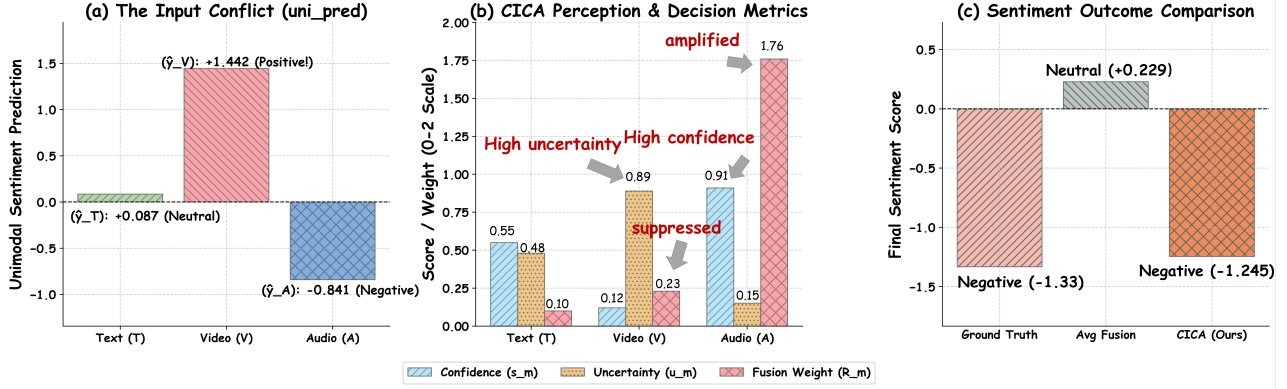


Figure 4. **Qualitative case: resolving cross-modal conflict.** (a) A misaligned input with conflicting visual (+1.442) and audio (−0.841) cues. (b) CICA’s internal perception metrics (s_m , u_m) and fusion weights (R_m) reveal how unreliable modalities are suppressed. (c) As a result, CICA predicts a correct negative sentiment ($\hat{y} = -1.245$), while the baseline fails.

These self-assessments guide the subsequent fusion.

In the decision step, CIF reduces the weight of unreliable signals: the visual modality receives **23%** of total weight, while audio dominates with **176%**. This adaptive reweighting suppresses misleading visual cues and stabilizes inference. Accordingly, the baseline model predicts a false positive (+0.229), whereas CICA outputs $\hat{y} = -1.245$, closely matching the ground truth. This case demonstrates that coupling perception and decision enables robust reasoning under strong cross-modal conflicts.

4.7. Visualization of Learned Representations

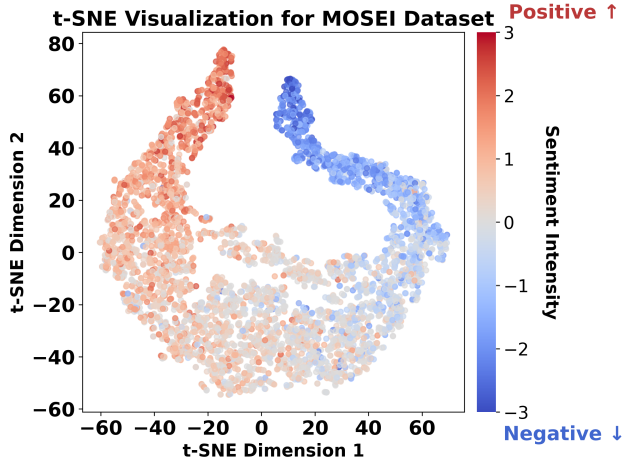


Figure 5. **t-SNE visualization on MOSEI.** Features from our model form a well-structured sentiment manifold with clear separation between positive (red) and negative (blue) samples. Full results across four datasets are provided in the Fig. 6.

We further visualize the learned multimodal embeddings to examine their discriminative organization. Using t-SNE on the MOSEI test set, as shown in Fig. 5, the fusion-layer

representations of CICA form a continuous and semantically ordered sentiment space, where features smoothly transition from negative (blue) to positive (red). Compared with the scattered and irregular baseline distribution, CICA produces compact clusters with clear sentiment gradients. Quantitative analysis in Appendix G supports this observation. For example, the Pos–Neg cluster distance reaches 58.46, reflecting strong separation and internal cohesion.

5. Conclusion

This work examines a central issue in multimodal sentiment analysis: current fusion models often fail when modalities conflict or contain noise or misalignment. We introduce **CICA**, a framework that connects perception and decision through a reliability-guided pipeline. In the **CAP** stage, each unimodal encoder outputs its feature embedding $\mathbf{H}_m$ and a reliability score s_m indicating signal quality. These scores are then used by the **CIF** module to guide fusion. The module first extracts intrinsic structural cues from each modality—token-level saliency ρ_m and local contextual patterns $\mathbf{N}_m$ —via the CIA mechanism, and then applies the confidence values s_m as continuous modulators of these representations. This coupling links a modality’s contribution to its estimated reliability, allowing fusion to account for both structural information and signal quality.

Experiments on MOSI, MOSEI, CH-SIMS, and CH-SIMSv2 show that CICA delivers strong benchmark performance and remains stable under noisy or missing modalities. Ablation studies indicate that the gains arise from the linkage between perception and decision rather than increased model capacity. Embedding confidence signals into the fusion step offers a practical and interpretable route toward more reliable multimodal reasoning models.

Acknowledgements

This work was supported by Sichuan Science and Technology Program (2025JDDQ0008, 2024NS-FJQ0035), and the Talents by Sichuan provincial Party Committee Organization Department.

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